Software Project Development with Agile Techniques

Introduction to the Unit

LECTURE ONE: Introduction to the Unit & Overview of Agile

GABRIEL KAPUMPE
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Over view of the course

By the end of the course students will be able to:

- Understand the background to Agile development;
- Understand the roles within an Agile development team;
- Understand the various Agile development techniques;
- Understand an Agile development lifecycle;
- Understand the principles associated with an Agile development approach;
- Be able to apply an Agile development techniques to a particular project scenario.

Introduction to Agile

Introduction to Agile Software Development
Agile Development Frameworks
The Importance of User-Centric Design

User-Centric Design

User Research & Personas
Design Thinking

Product Visioning

Crafting a Product Vision
Using Lean Canvas

Creating the Product Roadmap

Establishing the Product Backlog
Creating the Product Roadmap

The Scrum Framework

Roles & Responsibilities
Scrum Processes

Agile Architecture & Design

Prototypes & Wireframes
Agile Architecture Practices
Business Analysis & User Stories

Estimation & Planning

Estimation & Planning
Task Definition
Velocity

Agile Quality Management & Testing

Quality Management & Planning
Technical Practices & Standards
Software Testing Approaches

Continuous Delivery & DevOps

Continuous Integration, Delivery & Deployment
DevOps

Scaled Agile

Scaling Agile Frameworks

Today, we are looking at;

- Over view of the course
- What is Agile?
- Underpinning Themes
- The Waterfall?
- What is DSDM?
- Private Study Tasks

Assessment

This course is assessed by:

1. Class tests
2. One Semester assignment
3. Mid-Semester Examinations
4. Final Examinations

The Continuous Assessment and the Final Examinations worth **100%** of your overall mark for the course.

Exercise

What are the FIVE (5) words which you think describe Agile Development.

Underpinning Themes of Agile

Key points to consider are:

- User Involvement
- Iterative Development
- Incremental Development
- Flexibility to embrace change
- Prioritization
- Collaborative working

What is Agile?

a
set of defined principles
to be applied to the
development process.

What is Agile? - 1

There is a family of approaches:

- Dynamic Systems Development Method (DSDM)
- Lean Software Development (LSD)
- Feature Driven Development (FDD)
- Extreme Programming (XP)
- SCRUM
- Adaptive Software Development (ASD)
- Disciplined Agile Delivery (DAD), which includes Agile Modelling (AM)

What is Agile? - 2

Collectively, they address:

- Engineering best practices
- Leadership philosophy
- Project management framework
- Development framework

The Agile Manifesto - 1

"We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

- **Individuals** and **interactions** over processes and tools
- **Working software** over comprehensive documentation
- **Customer collaboration** over contract negotiation
- **Responding to change** over following a plan.

The Agile Manifesto – 2

That is, while there is value in the items on the right, **we value the items on the left more**.”

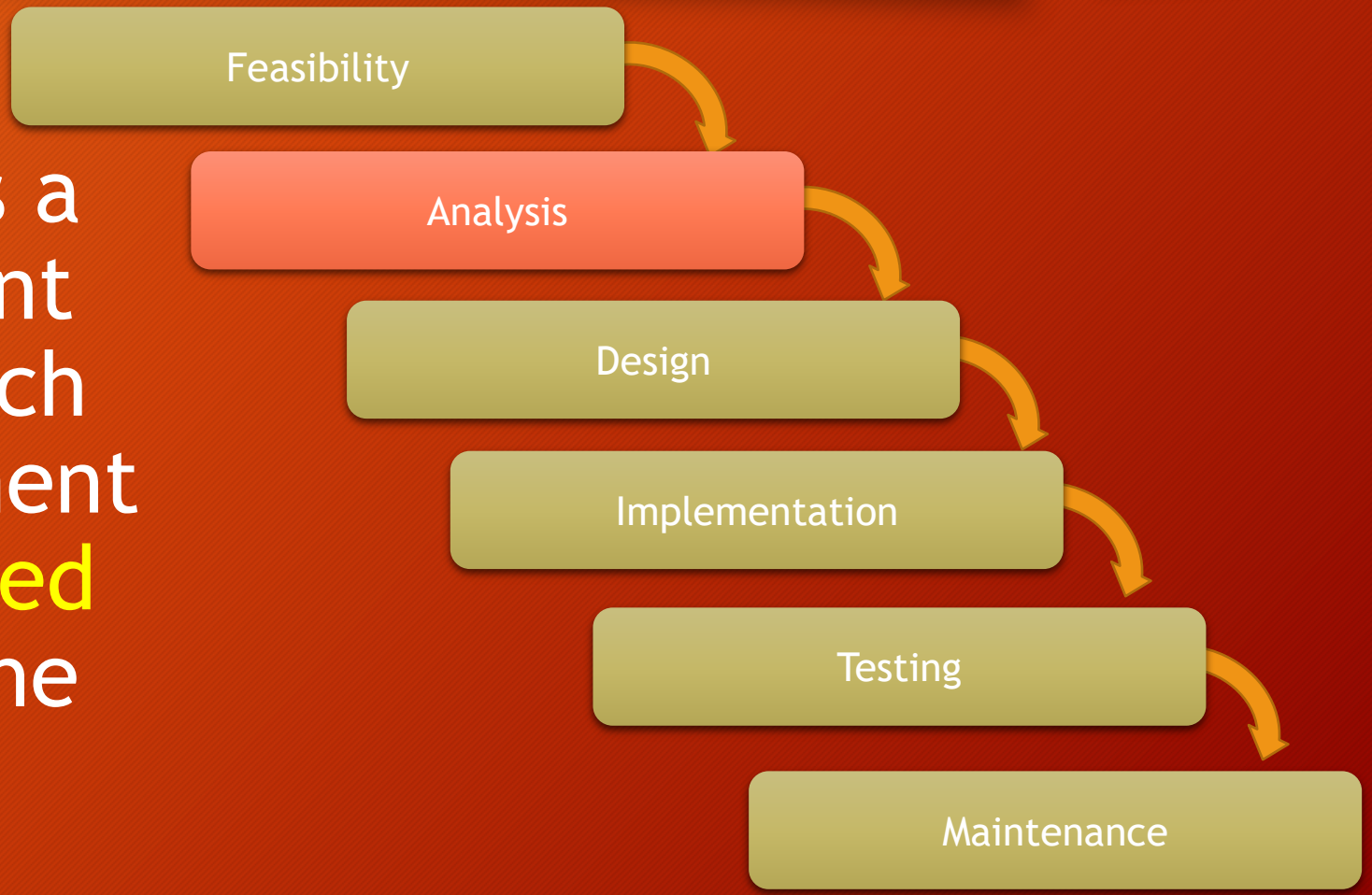
Source: www.agilemanifesto.org

Where does Agile come from?

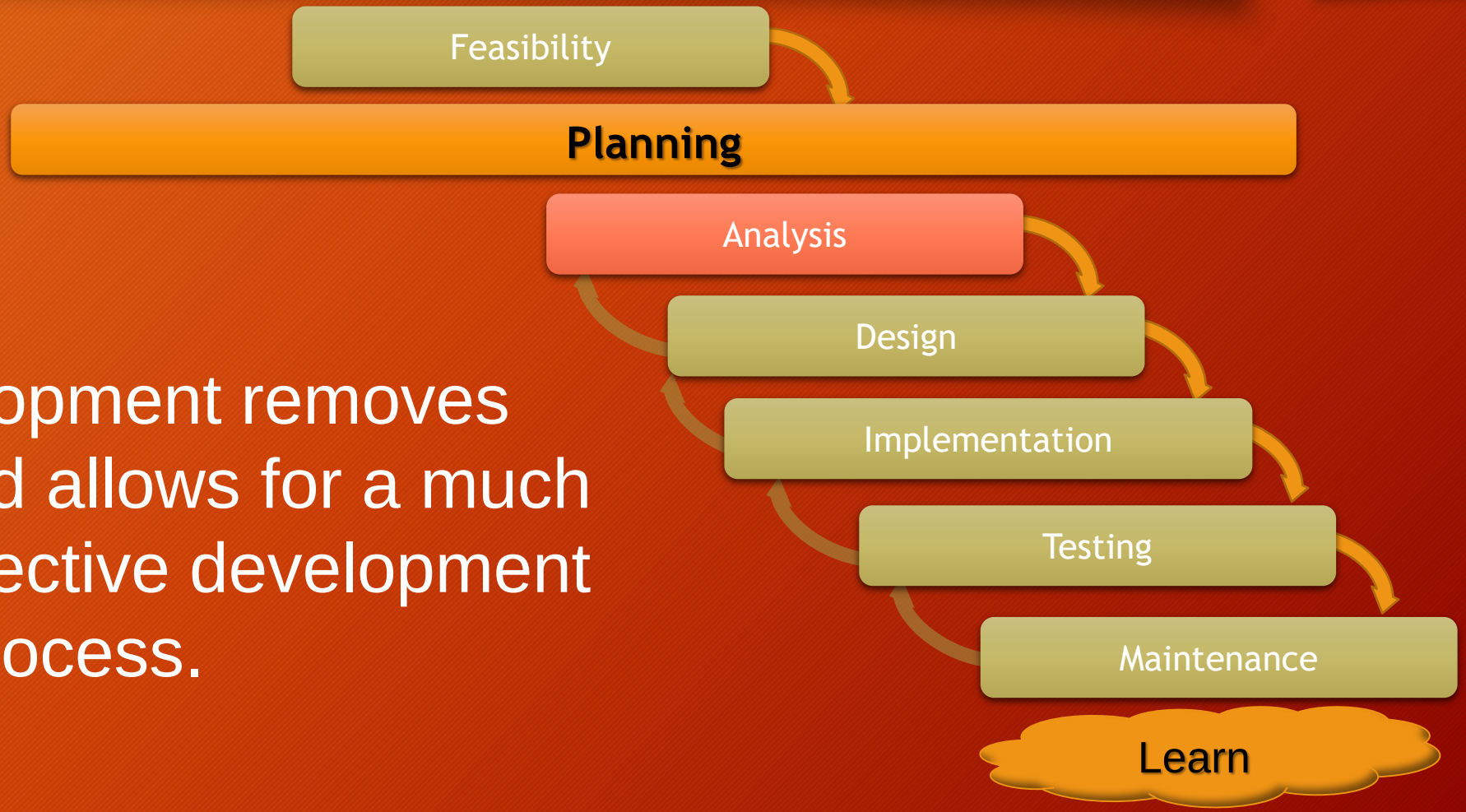
- In 1956, Herbert D. Benington described a process of development at a conference on advanced programming methods for digital computers.
- In 1970, Winston Royce wrote an article entitled: “Managing the Development of Large Software Systems”.
- These ideas formed the basis of what we know as the **The Waterfall Model**.
- **Agile** Development uses the stages from the Waterfall Model, but addresses its inadequacies.

What is The Waterfall Model?

- The Waterfall Model is a sequential development model. It insists on each stage of the development process to be completed before moving on to the next step.

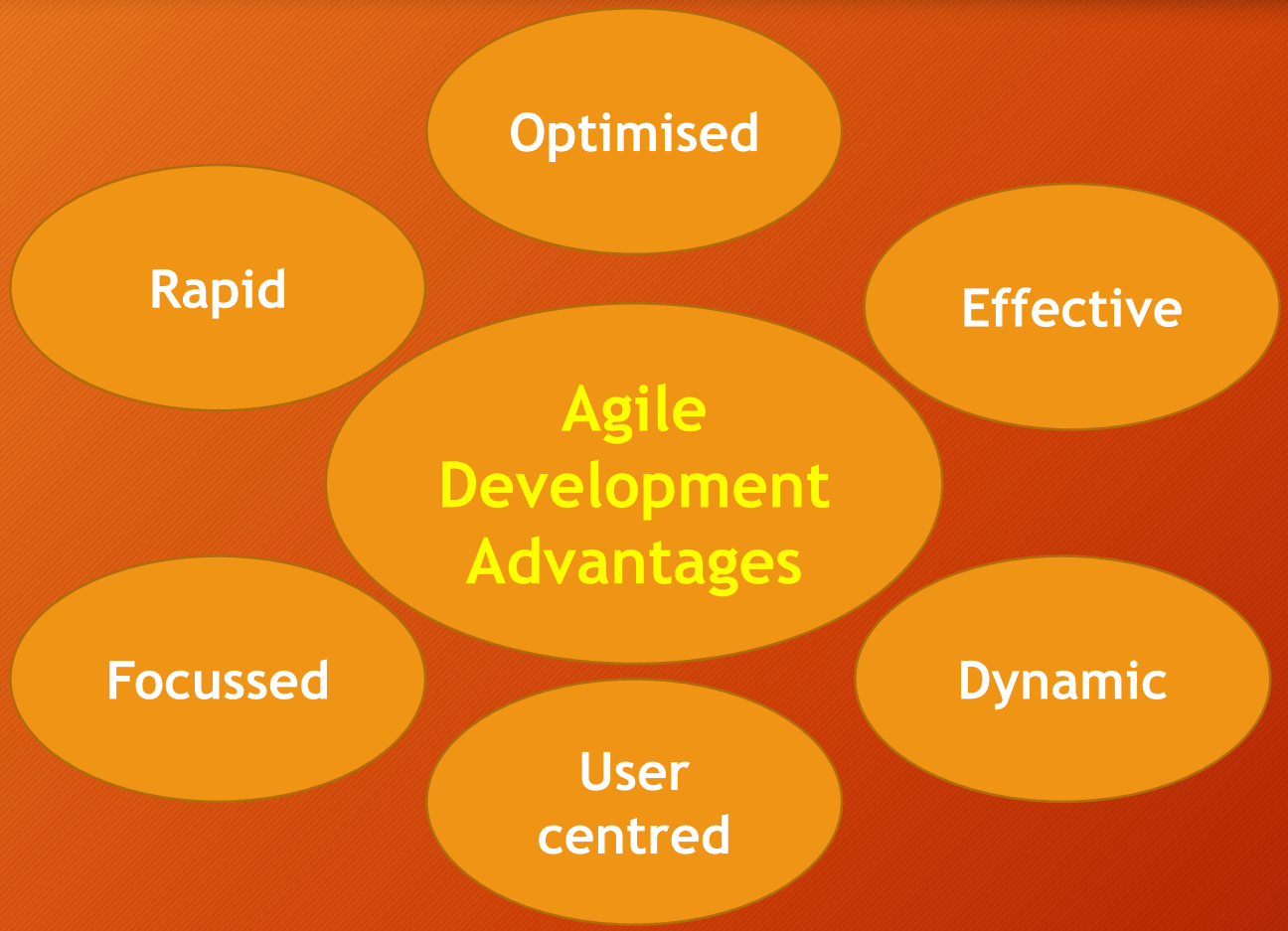


Agile is Iterative, Incremental, Evolutionary



Agile Development removes restrictions and allows for a much faster, cost effective development process.

Advantages of Agile Development



There are 10 steps to a successful project:

- **User involvement**
- **Executive management support**
- **Clear business objectives**
- **Optimizing scope**
- **Agile process**

Project management expertise
Financial management
Standard tools and Infrastructure
Skilled resources
Formal methodology

The five highlighted in bold apply to Agile Development, the others are resources that support the development process.

What is DSDM?

- **Dynamic Systems Development Method (DSDM) Agile Project Framework (2014 Onwards)**
- It is an Agile Project Delivery Framework that delivers the right solution at the right time. It is applicable to any kind of project. Key characteristics are:
    - Focused on business benefit;
    - On-time and in budget;
    - Quality and rigour;
    - Incremental and Collaborative.

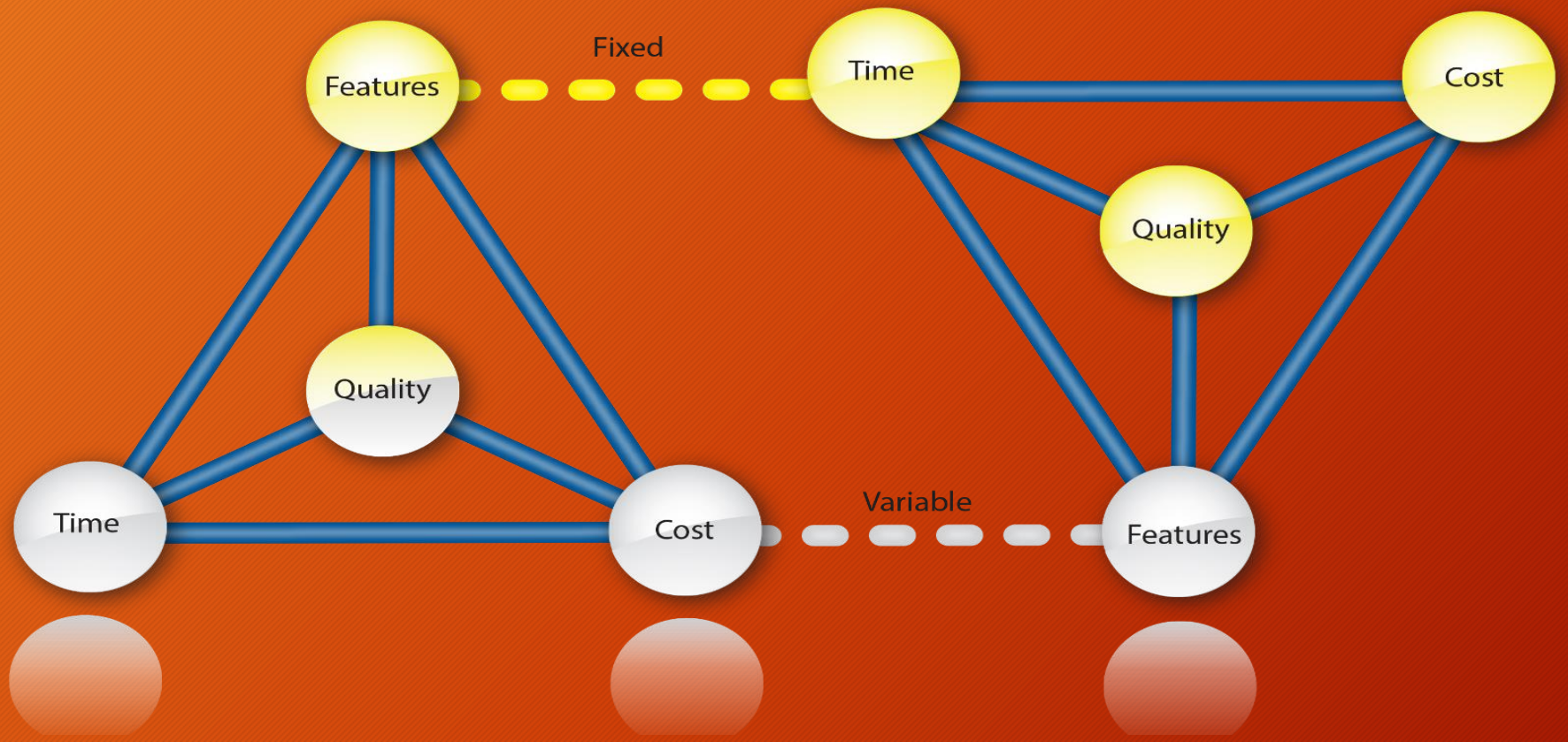
How does DSDM differ?

Traditional Approach

Traditional Approach

DSDM Approach

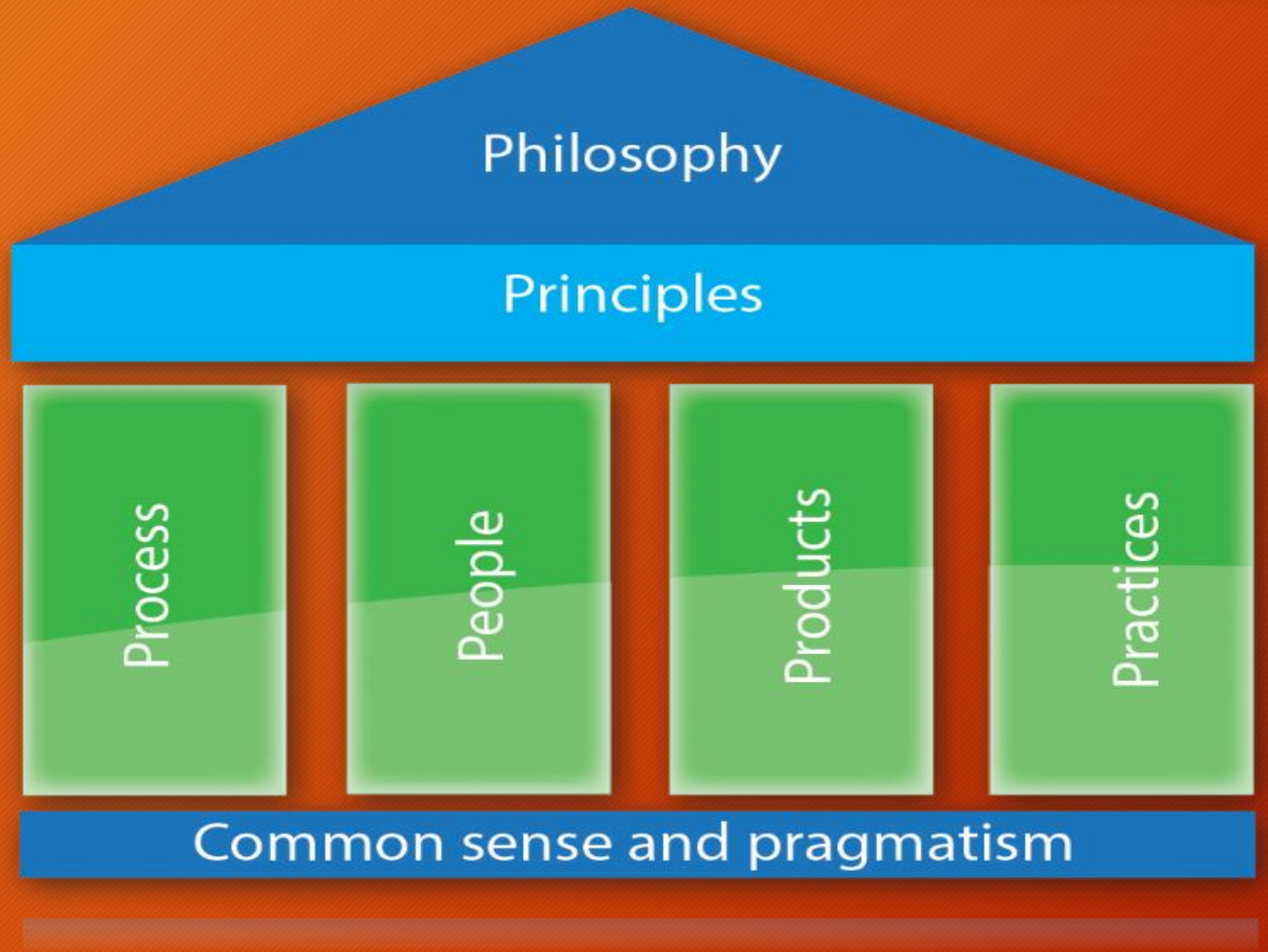
DSDM Approach



DSDM works with all the Agile principles and enhances the process by ensuring that strong foundations are planned at the start of the project. These foundations are built around three perspectives: Business, Solution and Management.

Source: Image from dsdm.org © 2016

The Structure of DSDM



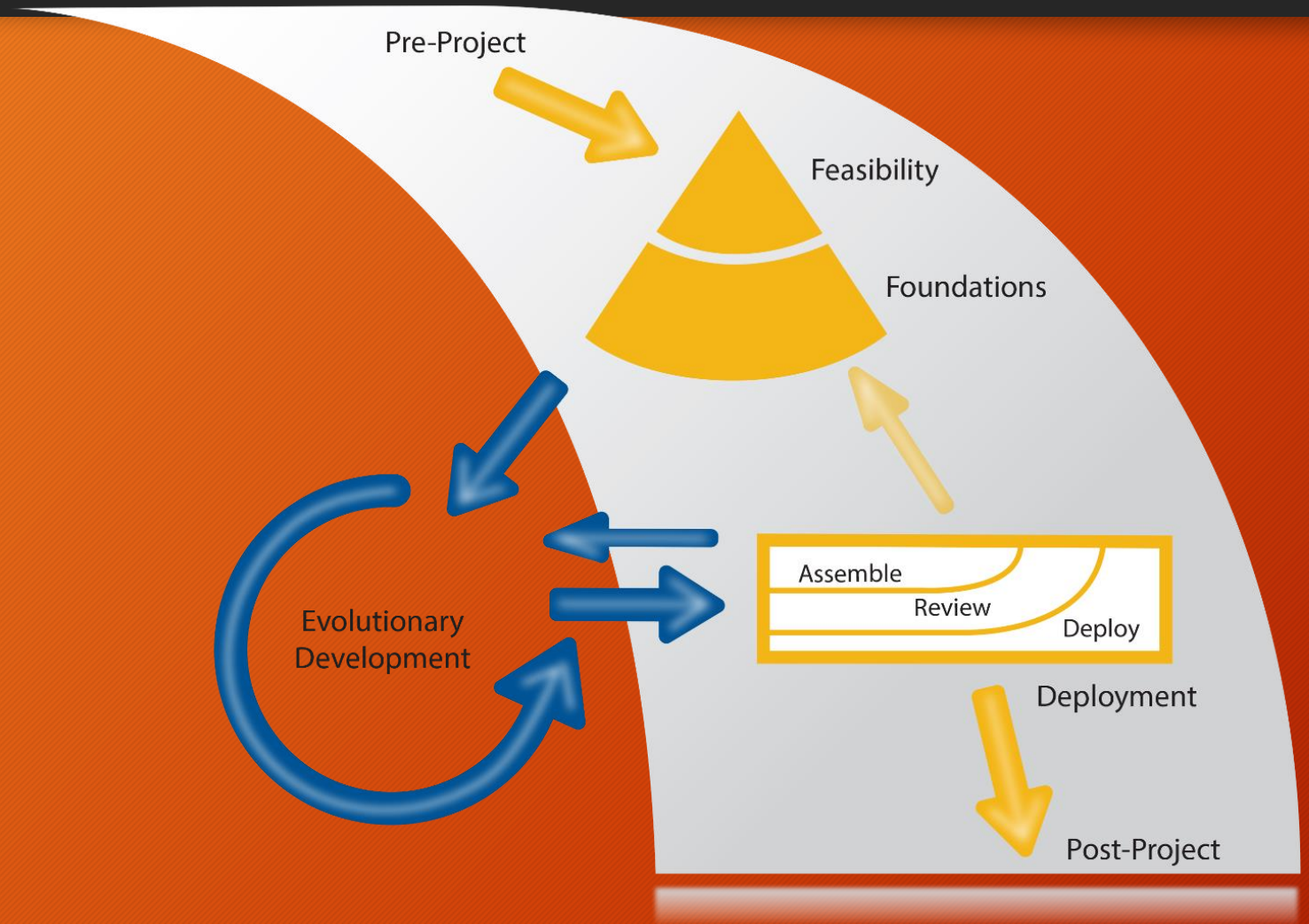
Four key components;

- Process
- People
- Products
- Practices

Each playing an equal role in ensuring that the project is delivered; on time, to cost, without compromising on quality.

Source: Image from dsdm.org © 2016

The Process



The Process is far more flexible than the Waterfall approach, and once the finely planned foundations are built, the project can evolve and adapt until the final project is deployed

Source: Image from dsdm.org © 2016

Summary of Lecture 1

- Today, we have:
  - Provided an introduction to the structure and content of the Software Project with Agile Techniques course;
  - Given the overall scope and coverage of the course;
  - Introduced the history and key concepts of Agile, in comparison to the waterfall approach;
  - Shown the structure and process of DSDM Agile Project Framework 2014

ANY QUESTIONS ???

Exercise – Action Research

1. Familiarize yourself with DSDM and give a case example of how it was applied in a real situation.

2. Focus on researching the following approaches; Scrum, Kanban, FDD, Lean and XP. These were introduced in this lecture. You should present a summary of **each approach in your own words in the next lecture**.